REPORT ON SALES PERFORMANCE ANALYSIS

Project Objective:

- To analyze overall sales performance by evaluating total sales, quantity sold, and product-wise contribution using Excel.
- To identify sales patterns and key drivers by comparing store-wise, category-wise, and year-wise sales trends.
- To support data-driven decision-making by creating an interactive dashboard with pivot tables, charts, and slicers.

Dataset Description:

The data set used in this project contains retail sales transaction data, capturing detailed information related to products, stores, and sales performance. It includes attributes such as Order Date, Product ID, Product Category, Brand, Store ID, Quantity Sold, Unit Price, Discount, and Total Sales Amount. The data set is structured in multiple tables and helps in analyzing sales performance over time, comparing products and stores, and creating interactive dashboards using Excel.

Column Description

Column Name	Description
Order Date	Date on which the sales transaction occurred
Order ID	Unique identifier for each sales order
Product ID	Unique code assigned to each product
Product Name	Name of the product sold
Category	Product category (e.g., Clothing, Electronics, etc.)
Brand	Brand of the product
Store ID	Unique identifier for each store
Store Name	Name of the retail store

Quantity Sold	Number of units sold in a transaction
Unit Price	Price of a single unit of the product
Discount	Discount applied to the product
Total/Total Amount	Final sales value calculated after applying discount

Data cleaning & transformation:

1. Data Import :

- The data set was imported into Microsoft Excel.
- All tables were converted into Excel Tables using **Ctrl + T** for better structure and analysis.

2. Remove Duplicates :

- Duplicate records were identified using **Data → Remove Duplicates**.
- Duplicates were removed based on key columns such as **Order ID** and **Product ID**.

3. Handle Missing Values

- Records with blank values in critical fields such as Order Date, Product ID, Store ID, and Quantity Sold were reviewed and removed to maintain data accuracy.
- Trim - remove spaces before after between words used in customer_name, product_name, Brand and remove non printable characters.
- **=IF(ISBLANK(G2),avg(G:G),G2)** used in unit price,cost quantity,total amount.

4. Data Types

- Order Date column was formatted as **Date**, Quantity Sold, Unit Price, Discount, and Total Sales columns were formatted as numeric values.**Currency format (₹)** was applied to sales-related columns.

5 Create Calculated Columns

- Customer_Dim column - In name column trim and cleaned =proper(B2) and for space **=TRIM(Substitute(B2,char(160),""))**, in country column united state of America replaced to USA using find and replace and in loyalty level column **=IF(H2,""No Loyalty",H2)**

- Prod_Dim column - product id column used find and replace PRODUCT TO PROD, in product brand column **=TRIM(B2)&" - "&TRIM(E2)**and for stock status **=if(G2=0,”out of stock”,if(G2<=100,”lower stock”,”in stock”))**
- Sales_fact column - Quantity column clean, trim =if(F2=””,O,F2) and in sales category **=IF(K2>10000,"High",IF(K2>5000,"Medium","Low")**. In this table Total Amount represents total revenue, as it is calculated after applying discounts. Therefore, Total Amount and Total Revenue are used interchangeably in the analysis.

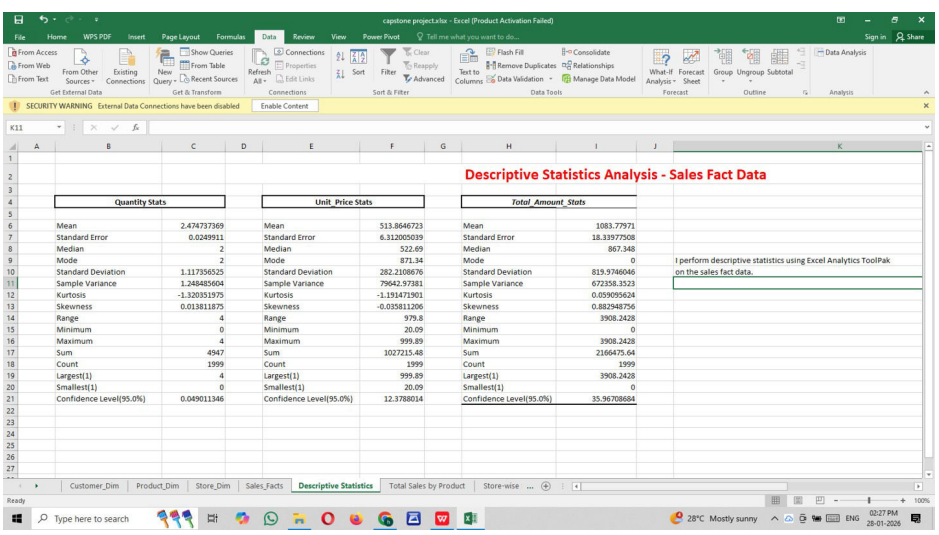
6 Formulas Used

- Total sales amount Revenue per order F2*G2
- Total Revenue F2*G2-I, for discount column blank =if(H2=””,0,H2)
- Total Revenue =SUM(H:H) and quantity sold =SUM(F:F)
- VLOOK UP was applied in sales fact to retrieve unit price from Product_Dim table using product id as look up key, ensuring data consistency =VLOOKUP(D2,product_dim!$A:$F,6,false,””)

7. Conditional Formatting

- Conditional formatting applied to data bars to quantity, unit price top 10% conditional formatting and for Total Revenue I used Color scales to identify high price product.

8. Descriptive Statistics



Descriptive Statistics Analysis - Sales Fact Data

Quantity Stats	
Mean	2.474737369
Standard Error	0.0249911
Median	2
Mode	2
Standard Deviation	1.117356525
Sample Variance	1.248485604
Kurtosis	-1.320351975
Skewness	0.013811875
Range	4
Minimum	0
Maximum	4
Sum	4947
Count	1999
Largest(1)	4
Smallest(1)	0
Confidence Level(95.0%)	0.049011346

Unit_Price Stats	
Mean	513.8646723
Standard Error	6.312005039
Median	522.69
Mode	871.34
Standard Deviation	282.2108676
Sample Variance	79642.97381
Kurtosis	-1.191471901
Skewness	-0.035811206
Range	979.8
Minimum	20.09
Maximum	999.89
Sum	1027215.48
Count	1999
Largest(1)	999.89
Smallest(1)	20.09
Confidence Level(95.0%)	12.3788014

Total_Amount_Stats	
Mean	1083.77971
Standard Error	18.33977508
Median	867.348
Mode	0
Standard Deviation	819.9746046
Sample Variance	672358.3523
Kurtosis	0.059095624
Skewness	0.882948756
Range	3908.2428
Minimum	0
Maximum	3908.2428
Sum	2166475.64
Count	1999
Largest(1)	3908.2428
Smallest(1)	0
Confidence Level(95.0%)	35.96708684

I perform descriptive statistics using on the sales fact data.

I perform Descriptive statistics using **Excel Analytics ToolPak** on the sales_fact. Descriptive statistics summarize the key characteristics of the sales data using measures such as total

revenue, quantity sold, and product count. Its provide a clear overview of overall sales performance and distribution across products, stores, and time periods.

RELATIONSHIP PIVOT TABLE & PIVOT CHART

1. Total Sales by Product

The *Product Dimension* table is connected to the *Sales Fact* table using Product ID as the primary–foreign key relationship. Because of this relationship, the Pivot Table aggregates Total Sales (Revenue) from the Sales Fact table and displays it product-wise, ensuring each sale is correctly assigned to its product. For pivot chart I preferred Column Chart.

2. Quantity Sold by Product

Quantity Sold is stored in the *Sales Fact* table, while Product Name is stored in the *Product Dimension* table. Using the Product ID relationship, the Pivot Table sums the Quantity Sold for each product, allowing accurate analysis of product demand. For pivot chart I prefered bar chart.

3. Store-wise Sales Performance

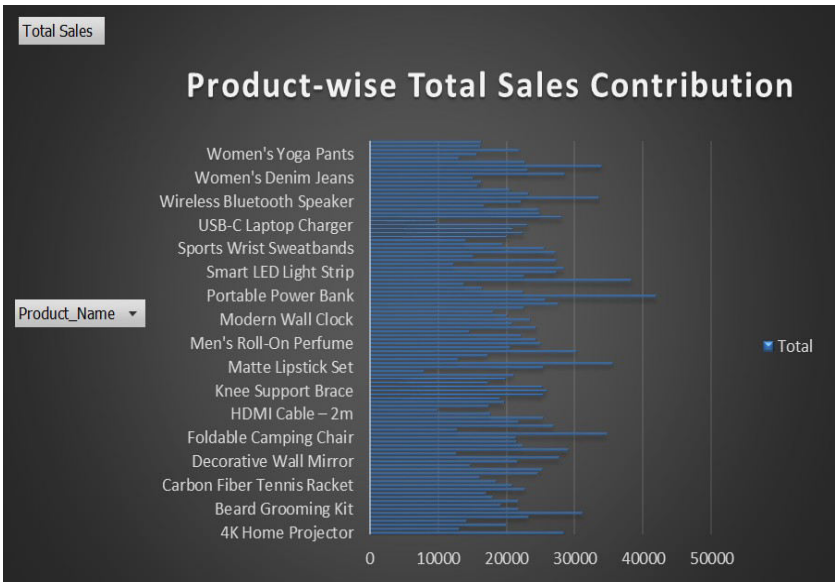
The *Store Dimension* table is linked to the *Sales Fact* table using Store ID as the common key. This relationship enables the Pivot Table to group Total Sales by store, making it possible to compare sales performance across different store locations. For pivot chart I preferred Column Chart.

4. Sales Trend (Time-based Analysis)

The *Order Date* from the *Sales Fact* table is used for time analysis, where dates are grouped by year or month in the Pivot Table. This allows sales values to be aggregated over time, revealing sales trends and growth patterns across different periods.

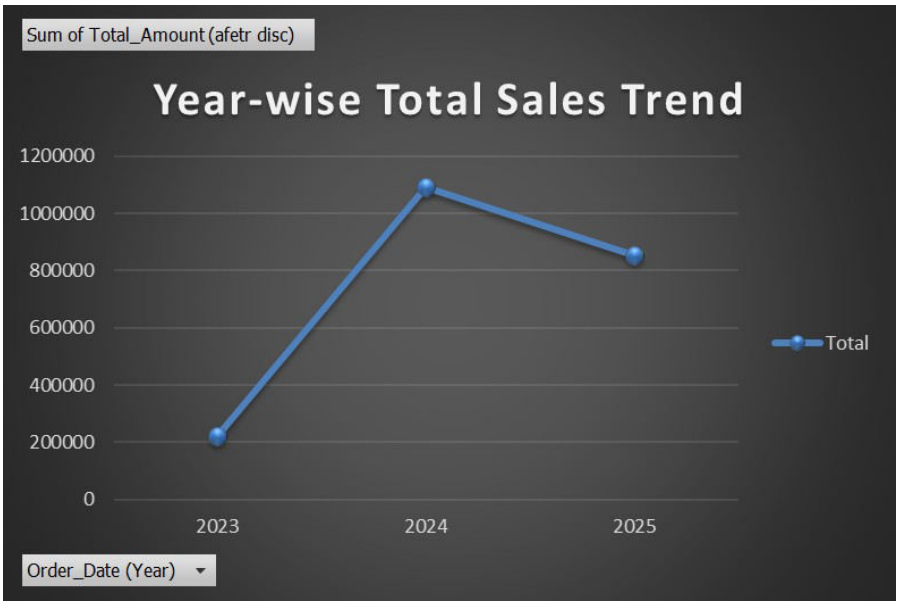
Visualization and Insights Arrived with screenshot of visualization

Descriptive analysis: This visualization shows the total sales contribution of each product. It helps identify top-selling and low-performing products based on historical sales data.

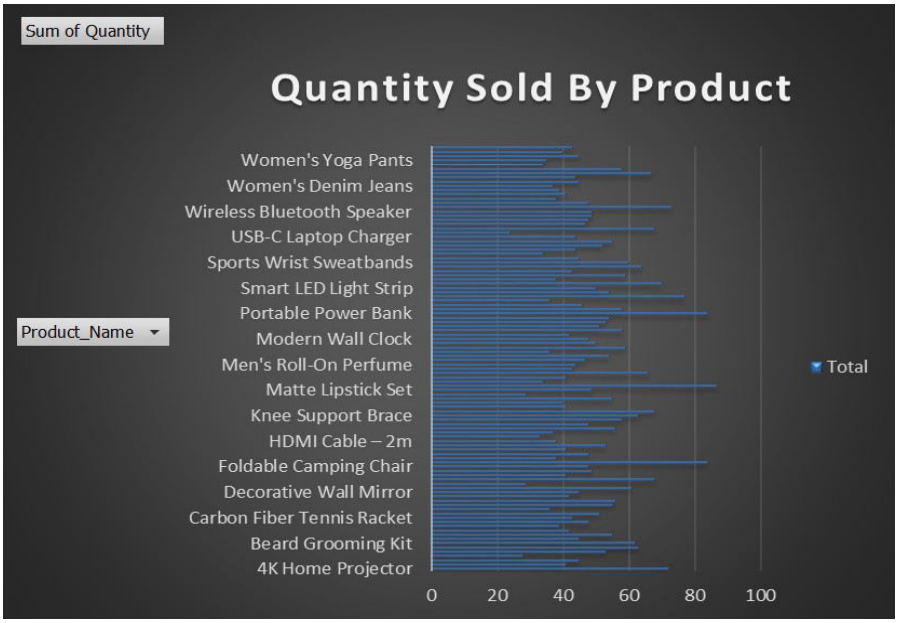


Diagnostic Analysis : This chart explains why sales differ across stores by comparing store-wise performance. It highlights stores contributing higher or lower sales, helping identify performance gaps.

3. Predictive Analysis: This visualization shows sales trends over time based on past data. The trend helps predict future sales patterns and seasonal demand behavior.



4. Prescriptive Analysis: This analysis helps suggest actions by identifying high-demand products and categories. The insights support better inventory planning and focused sales strategies.



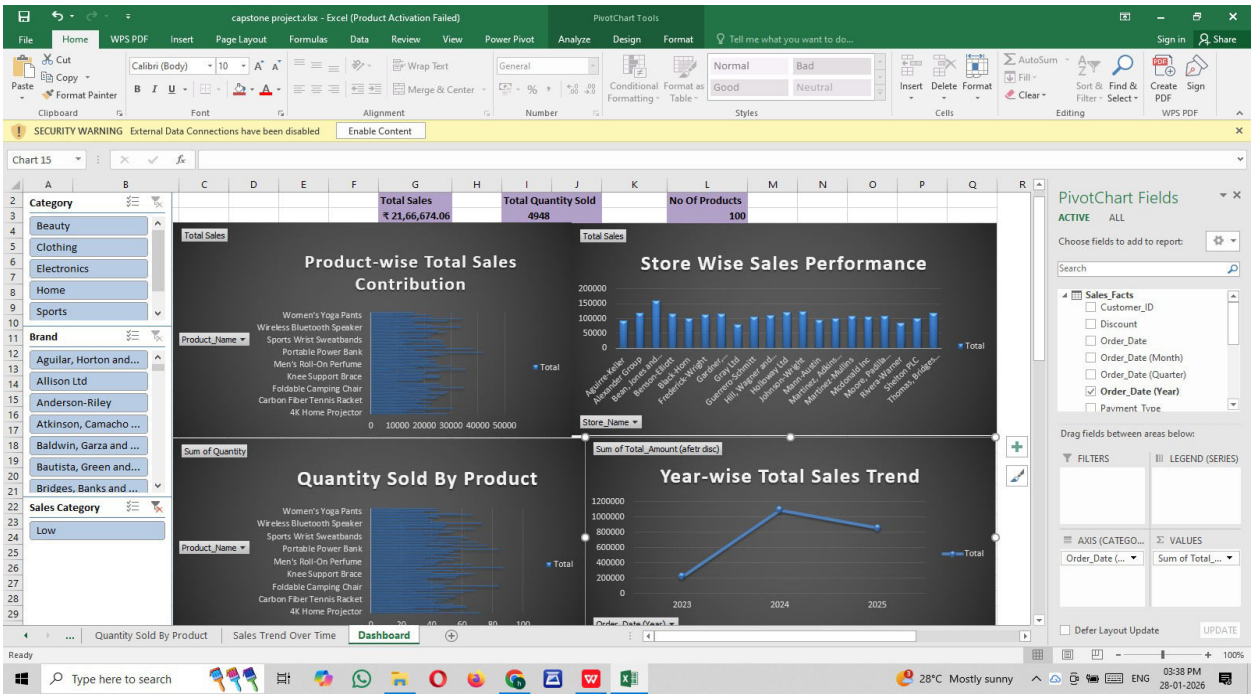
DASHBOARD CREATION This dashboard provides an integrated view of sales performance across products, stores, and time periods. It helps transform raw sales data into meaningful insights that support informed business decisions. Key performance indicators such as **Total Revenue**, **Quantity Sold**, and product and store contributions are highlighted for quick understanding.

The **Product-wise Sales** and **Quantity Sold by Product** charts help identify top-performing and low-performing products, enabling better inventory and pricing decisions. The **Store-wise Sales**

Performance visualization compares sales across different store locations, making it easier to evaluate store efficiency and recognize performance gaps.

The **Sales Trend** chart presents how sales change over time, revealing growth patterns and seasonal variations. This supports trend-based forecasting and planning. Interactive **slicers** for category and brand allow users to filter data dynamically, enhancing usability and enabling focused analysis without altering the underlying dataset.

Overall, the dashboard supports descriptive and diagnostic analysis while providing a foundation for predictive and prescriptive insights, making it an effective tool for monitoring and improving sales performance.



Conclusion:

Based on the sales trend analysis and historical patterns, overall sales are expected to continue following a similar growth trend in the coming periods. Products and categories that consistently show high sales performance are likely to remain strong contributors, while low-performing products may continue to underperform unless strategic changes are made.